

Skills Summary

Composing Functions from Symbolic Rules

Use this reference while completing your worksheet or when studying.

Skill 1 — Composing at a number

Notation: $(f \circ g)(a) = f(g(a))$. The function written *closest to the input* runs first. Two machines in a row: work out the inner output, then feed it into the outer rule.

Example: For $f(x) = 4x - 1$ and $g(x) = x^2$, find $(f \circ g)(3)$.

Step 1. Only one number has to travel through the chain, so evaluate the inner machine at the input and pass its output on — no algebra is needed.

Step 2. $g(3) = 3^2 = 9$, then $f(9) = 4(9) - 1 = 36 - 1$.

$\Rightarrow$ **35**

Try it: For $f(x) = x + 7$ and $g(x) = 2x$, find $(f \circ g)(5)$.

check: 17

Skill 2 — Composing into a new rule

Method: replace every x in the *outer* rule by the whole *inner* rule, in brackets, then simplify. The brackets are what protect a squaring or a negative sign from being applied to only part of the inner rule.

Example: For $f(x) = x^2 - 2$ and $g(x) = x + 3$, find $(f \circ g)(x)$.

Step 1. The whole inner rule is the input to f , so it goes in wherever f 's formula shows an x — bracketed, because f squares its input.

Step 2. $f(g(x)) = (x + 3)^2 - 2 = x^2 + 6x + 9 - 2$.

$\Rightarrow$ **$x^2 + 6x + 7$**

Try it: For $f(x) = 5x + 2$ and $g(x) = x - 4$, find $(f \circ g)(x)$.

check: $5x - 18$

Skill 3 — The two orders are different

Fact: $f \circ g$ and $g \circ f$ are usually *not* the same function. Composition is not like multiplication: swapping the order changes which operation happens first, and that changes the rule. Always check which function the notation puts closest to the input.

Example: For $f(x) = 3x - 2$ and $g(x) = x + 5$, build both orders.

Step 1. Each order is a separate substitution, so build them one at a time and compare the finished rules rather than guessing.

Step 2. $f(g(x)) = 3(x + 5) - 2 = 3x + 15 - 2$, while $g(f(x)) = (3x - 2) + 5$.

$f \circ g$: **$3x + 13$** $g \circ f$: **$3x + 3$**

Try it: For $f(x) = 4x + 1$ and $g(x) = x - 2$, build both orders.

1 - $x \mapsto f \circ g$, 2 - $x \mapsto g \circ f$:check

Skill 4 — Solving with a composition

Method: build the composed rule *first*, so the equation becomes an ordinary one in x , then solve it as usual. Finish by running your answer through both machines to check it.

Example: For $f(x) = 3x + 1$ and $g(x) = x - 2$, solve $(f \circ g)(x) = 10$.

Step 1. Nothing can be solved while the unknown is still inside two functions, so compose into a single rule before touching the equation.

Step 2. $f(g(x)) = 3(x - 2) + 1 = 3x - 5$, so $3x - 5 = 10$ and $3x = 15$. Check: $g(5) = 3$ and $f(3) = 10$.

$$\Rightarrow x = 5$$

Try it: For $f(x) = 2x - 3$ and $g(x) = x + 6$, solve $(f \circ g)(x) = 21$.

9 = x :check